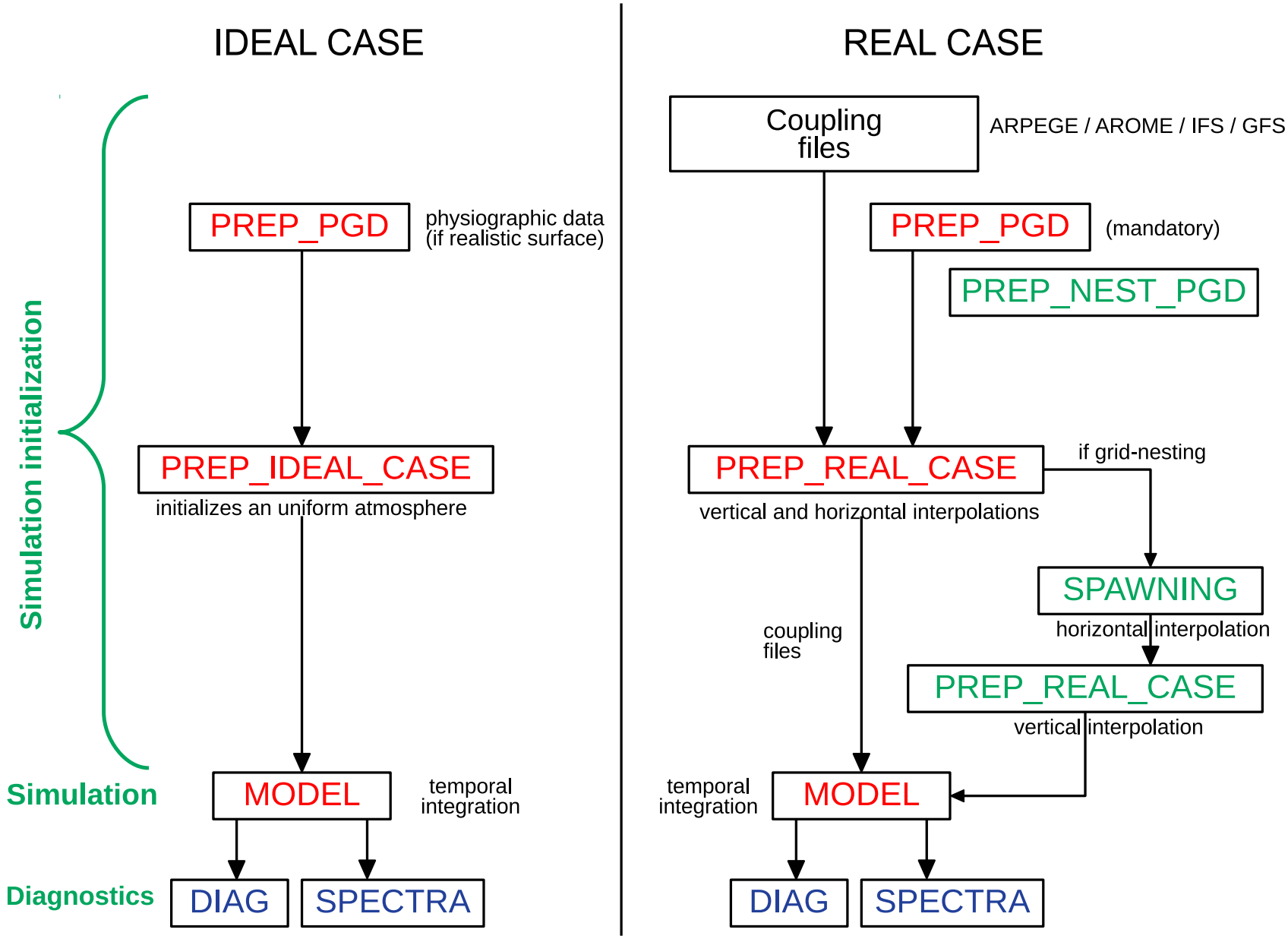
Ideal case

MesoNH Tutorial Class 2-5 November 2026


IDEAL CASE
REAL CASE
Simulation initialization
PREP_PGD
physiographic data
(if realistic surface)
PREP_IDEAL_CASE
initializes an uniform atmosphere
Simulation
MODEL
temporal
integration
Diagnostics
DIAG
SPECTRA
Coupling
files
ARPEGE / AROME / IFS / GFS
PREP_PGD
(mandatory)
PREP_NEST_PGD
PREP_REAL_CASE
vertical and horizontal interpolations
if grid-nesting
SPAWNING
horizontal interpolation
coupling
files
PREP_REAL_CASE
vertical interpolation
temporal
integration
MODEL
DIAG
SPECTRA

Interests

- theorical study :
  from initial idealised fields
  in 1D,2D,3D configuration
  in cartesian geometry or conformal projection
- tests of validation

How does it work ?

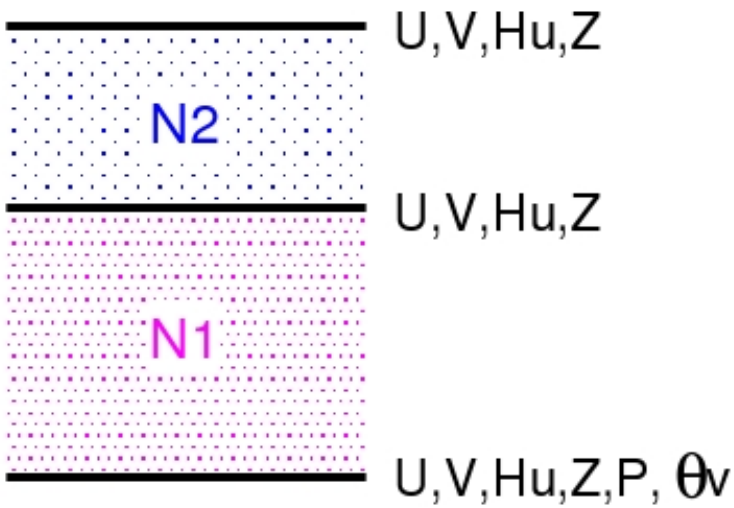
The user

1. specifies an uniform atmosphere from a profile
2. can add a perturbation
3. initializes the surface fields : idealised or realistic

Uniform state

Vertical profile defined by :

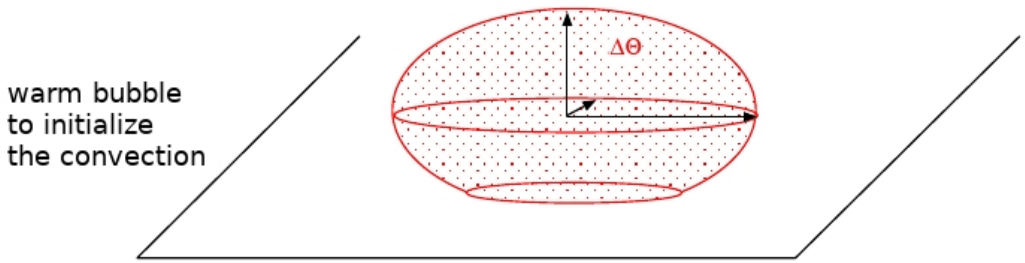
- some layers with constant Brunt-Väisälä frequency, wind and humidity at the interfaces : CIDEAL='CSTN'



- or data from a radio-sounding : CIDEAL='RSOU'

Analytical perturbations

- on thermodynamic fields



- on potential temperature : white noise for LES simulations
- on non-divergent components of the wind

namelist &NAM_PERT_PRE
to modify according to the user's needs : routine set_perturb.f90

Forcing terms

1D chronological serie to impose a large-scale environment :

- constant translation of the domain, geostrophic forcing (u_{frc}, v_{frc})
- vertical transport : w_{frc}
- horizontal transport : $\left.\frac{\partial \theta}{\partial x}\right|_{frc}$, $\left.\frac{\partial \theta}{\partial y}\right|_{frc}$, $\left.\frac{\partial r_v}{\partial x}\right|_{frc}$, $\left.\frac{\partial r_v}{\partial y}\right|_{frc}$
- newtonian force : u_{frc}, v_{frc}, θ_{frc}, r_{vfrc}
- tendency : $\left.\frac{\partial \theta}{\partial t}\right|_{frc}$, $\left.\frac{\partial r_v}{\partial t}\right|_{frc}$, $\left.\frac{\partial u}{\partial t}\right|_{frc}$, $\left.\frac{\partial v}{\partial t}\right|_{frc}$

Forcing terms

Documentation : user's guide > free-format part > the forced version

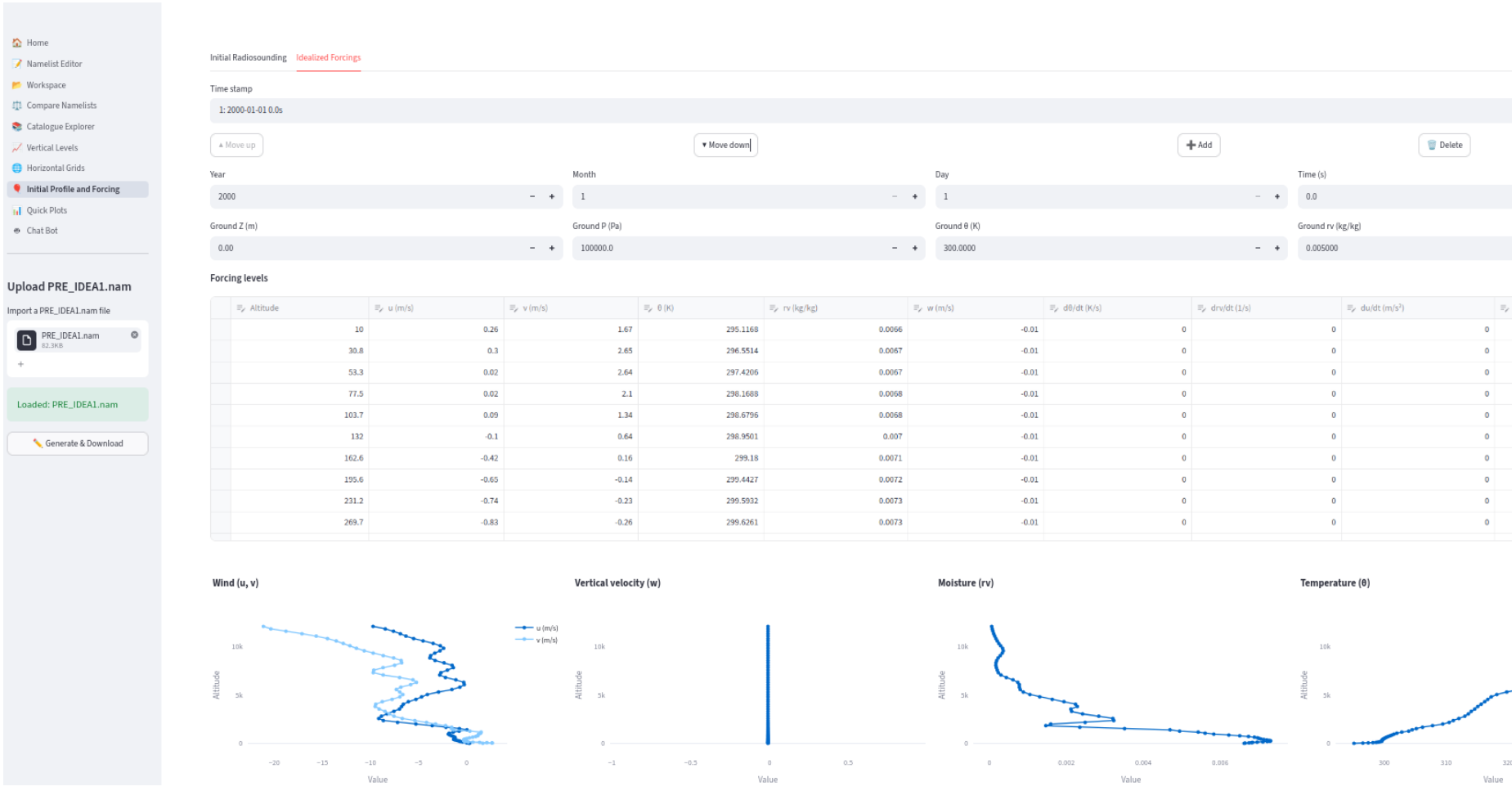
Note

Example of free part of `PRE_IDEA1.nam` :

```
ZFRC
1
1983 07 01 0.
0
1000000
284.5
0.008
6
   5. -7.0 0.0 281.10 0.00540 -0.00000 0. 0. 0. 0.
  15. -7.0 0.0 281.10 0.00540 -0.00000 0. 0. 0. 0.
1095. -7.0 0.0 280.75 0.00540 -0.00300 0. 0. 0. 0.
1145. -7.0 0.0 290.60 0.00190 -0.00300 0. 0. 0. 0.
3000. -7.0 0.0 304.15 0.00190 -0.00300 0. 0. 0. 0.
9000. -7.0 0.0 346.15 0.00190 -0.00300 0. 0. 0. 0.
```

Forcing terms

mesonh.streamlit.app/Initial_radiosoundings_forcing



Surface fields

- Orography (namelist &NAM_CONF_PRE) :
  - idealised : flat (CZS='FLAT'), sinusoidal (CZS='SINE') or bell-shaped (CZS='BELL')
  - real : discretised (CZS='DATA') or read in a MesoNH PGD file
- Physiographic data (land-sea mask, type of cover SST, LAI, lay, sand...)
  - idealised : uniform
  - realistic : MesoNH PGD
- Prognostic fields (ground temperatures `TG_SURF, _ROOT _DEEP`, water contents `WG_SURF, _ROOT _DEEP`)
  - idealised : uniform
  - realistic : from a file from an operational model or from a MesoNH file

PREP_IDEAL_CASE

PREP_IDEAL_CASE

- computation of the horizontal grid (cartesian or conformal) and vertical one
- interpolation of the atmospheric fields with respect to :
  - hydrostatic balance
  - possible geostrophic balance
  - correction of u, v, w by the pressure solver

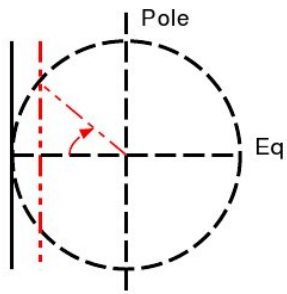
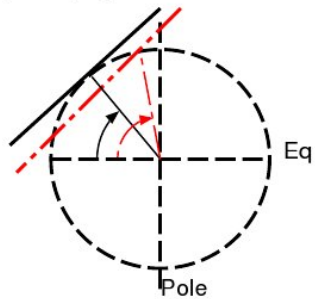
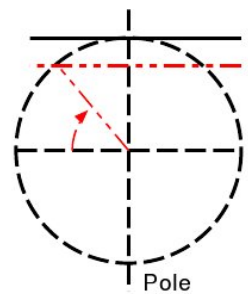
  to verify the ground condition and the anelastic constraint
- computation of the profiles of the reference state (anelastic approximation)
- writing in a NetCDF file of all the fields necessary for a simulation (prognostic : u, v, w, θ, r_v diagnostic P, surface fields)

Choice of the horizontal grid

- number of points : $\texttt{IMAX} = 2^p * 3^q * 5^r$, $\texttt{JMAX} = 2^s * 3^t * 5^u$
- meshes : `DELTAX`, `DELTAY` (in meters)

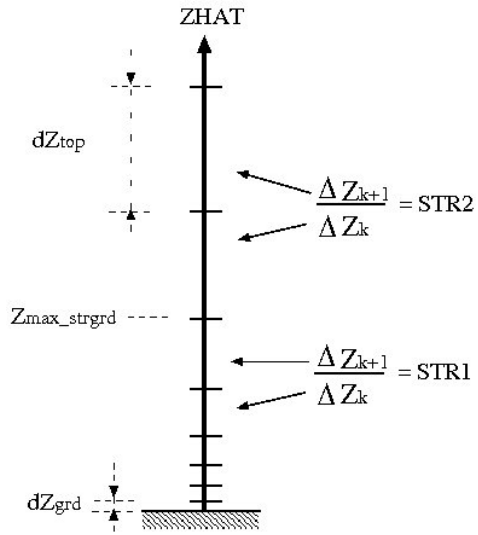
1 2 3 4 5 6 8 9 10 12 15 16 18 20 24 25 27 30 32 36 40 45 48 50 54 60
64 72 75 80 81 90 96 100 108 120 125 128 135 144 150 160 162 180 192
200 216 225 240 243 250 256 270 288 300 320 324 360 375 384 400 405
432 450 480 486 500 512 540 576 600 625 640 648 675 720 729 750 768
800 810 864 900 960 972 1000 1024 1080 1125 1152 1200 1215 1250
1280 1296 1350 1440 1458 1500 1536 1600 1620 1728 1800 1875 1920
1944 2000 2025 2048 2160 2187 2250 2304 2400 2430 2500 2560 2592
2700 2880 2916 3000 3072 3125 3200 3240 3375 3456 3600 3645 3750
3840 3888 4000 4050 4096 4320 4374 4500 4608 4800 4860 5000 5120
5184 5400 5625 5760 5832 6000 6075 6144 6250 6400 6480 6561 6750
6912 7200 7290 7500 7680 7776 8000 8100 8192 8640 8748 9000 9216
9375 9600 9720 10000 10125 10240 10368 10800 10935 11250 11520
11664 12000 12150 12288 12500 etc

- Domain geometry
  - a **cartesian geometry** (`LCARTESIAN=T`) : Earth sphericity isn't take in account (for small scale phenomena),
  - **conformal projection**.

	Mercator	Lambert Conforme	Stéréographique Polaire
	(cylindrique)	(conique)	(plane)
			
	`XRPK=0`	`0<XRPK<1`	`XRPK=1`
		`(-1<XRPK<0)`	`(XRPK=-1)`
tangent	`(XLAT0=0)`	`(0<XLAT0<90)`	`(XLAT0=90)`
sécant	`(0<XLAT0<90)`	`(0<XLAT0<90)`	`(0<XLAT0<90)`
domaines :	lat. tropicales	lat. tempérées	lat. polaires

Choice of the vertical grid

number of points : `KMAX`
discretisation :

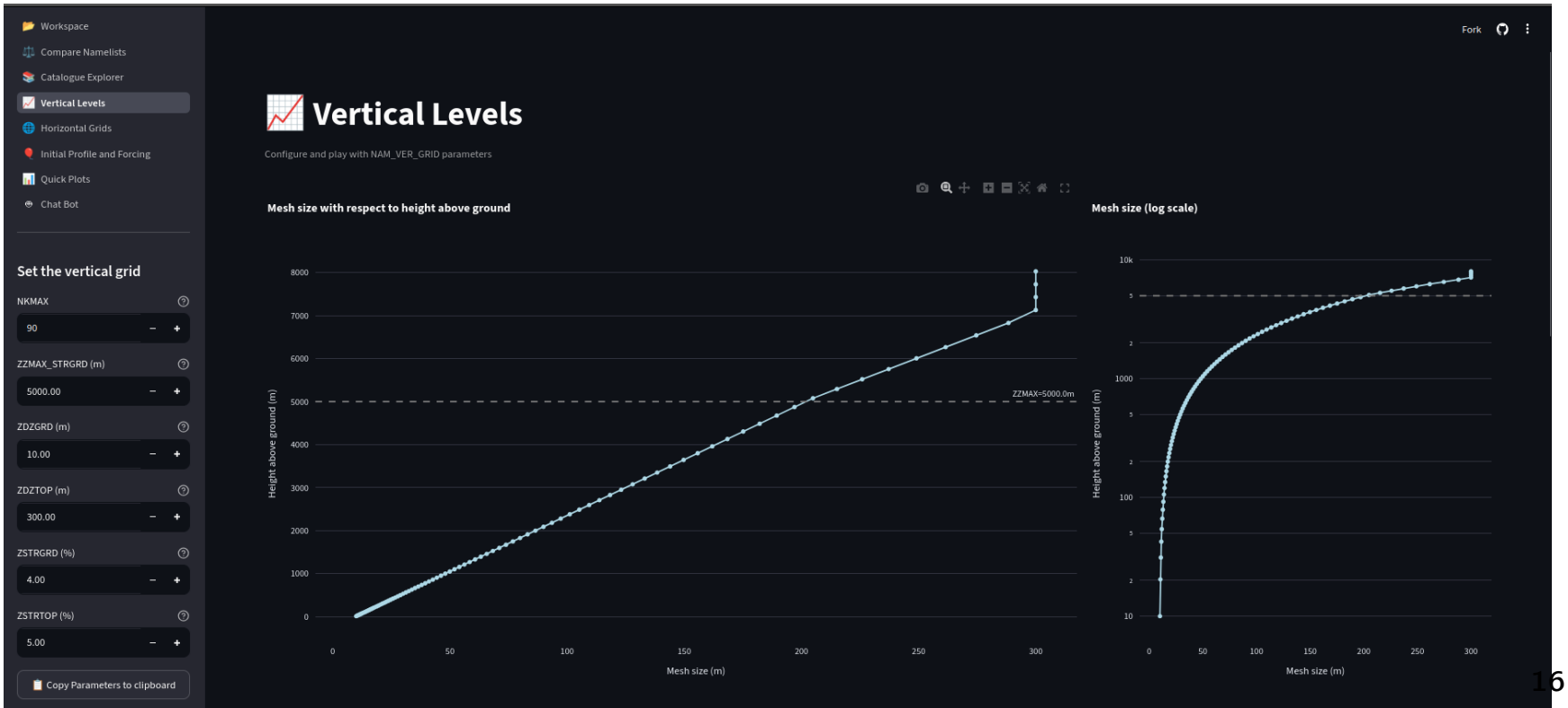


- manual : list of levels `ZHAT(1)`, `ZHAT(2)`,..., `ZHAT(NKMAX)`
- or analytical (logarithmic with z) :
  Computation of `ZHAT` from :
  - ΔZ at ground (ZDZGRD)
  - ΔZ at top (ZDZTOP)
  - STRetching ($0 \leq \Delta Z^{k+1}/\Delta Z^{k} < 10\%$) :
    ZSTRGRD for $z \in [0, ZZMAX_STRGRD]$
    ZSTRTOP for $z > ZZMAX_STRGRD$

Vertical grid

mesonh.streamlit.app/Vertical_Levels

```
&NAM_VER_GRID NKMAX=30,  number of points in Z
              YZGRID_TYPE='FUNCTN',  FUNCTN or MANUAL
              ZDZGRD=60., ZDZTOP=700.,  ΔZ at ground/at top
              ZZMAX_STRGRD=2500.,  Height for streching change
              ZSTRGRD=9.,ZSTRTOP=7.  / streching at ground/at top
```



PRE_IDEA1.nam

```
&NAM_DIMn_PRE NIMAX=30, NJMAX=30 /
```
number of points in i and j

```
&NAM_CONF_PRE LCARTESIAN=.TRUE.,
              CEQNSYS='LHE',
              CZS='BELL',
              CIDEAL='CSTN',
              LPERTURB=.T. /
```
cartesian geometry
choice of equation system
choice of orography
type of initialisation of the atmosphere
add perturbation

```
&NAM_GRIDH_PRE XDELTAX=4 .E3, XDELTAY=4.E3,
               XHMAX=1000., XAX=10.E3, XAY=10.E3
               NIZS=16, NJZS=16/
```
mesh in X and Y
define the bell

```
&NAM_VER_GRID NKMAX=24,
              ZDZGRD=500., ZDZTOP=500. /
```
number of vertical levels
ΔZ at top and at ground

```
&NAM_PERT_PRE CPERT_KIND='TH',
              XAMPLITH=1.5,
              XCENTERZ=1500.,
size of perturbationXRADX=10.E3,XRADY=10.E3, XRADZ=600. /
```
type of perturbation
max amplitude for θ
height for $\Delta\theta_{max}$

```
&NAM_LUNITn CINIFILE='BILAN.1' /
```
name of output file

```
&NAM_LBCn_PRE CLBCX = 2*"OPEN",
              CLBCY = 2*"OPEN" /
```
lateral boundary conditions

```
&NAM_VPROF_PRE CTYPELOC='IJGRID',
               NILOC=10, NJLOC=10,
               LGEOSBAL=.FALSE. /
```
variables to defined the profile
localisation of vertical profile
no geostrophic balance

```
CSTN
2000 01 01 0.
2
285.
100000.
0. 20000.
10. 10.
0. 0.
0. 0.
0.01
```

date
number of levels
θ_V *at ground*
P at ground
height at all levels
zonal wind component at all levels
meridian wind component at all levels
relative humidity at all levels
Brünt-Vaisala frequency at all layers

MODEL

EXSEG1.nam

```
&NAM_LUNITn CINIFILE = "BILAN.1" /initial file

&NAM_DYNn XTSTEP = 40., time step (s)
          CPRESOPT = "CRESI", LITRADJ = T,pressure solver
horizontal LHORELAX_UVWTH = T, LHORELAX_RV = T,
relaxation NRIMX = 6, NRIMY = 6, XRIMKMAX = 0.0005,
          LVE_RELAX = T,vertical relaxation
          /

&NAM_ADVn CUVW_ADV_SCHEME = "WENO_K" ,advection scheme for U,V
          NWENO_ORDER=5 , CTEMP_SCHEME='RK53'
          CMET_ADV_SCHEME = "PPM_01", /for θ,r,TKE

&NAM_PARAMn CCLOUD = "ICE3",CTURB = "TKEL",
             microphysic      turbulence
            CDCONV = "KAFR",CSCONV = "EDKF",
             deep convection  shallow convection
            CRAD = "ECMW" /radiative scheme
```

```
&NAM_PARAM_KAFRn XDTCONV = 300., NICE = 1,
parameters for convection KAFR  LDIAGCONV = F /

&NAM_PARAM_RADn XDTRAD = 1800., XDTRAD_CLONLY = 900.,
parameters for radiative scheme LCLEAR_SKY = F /

&NAM_TURBn CTURBDIM = "1DIM", CTURBLEN = "BL89",
turbulence parameters       LTURB_FLX = F, LTURB_DIAG = F /
&NAM_NEBn LSUBG_COND = T /

&NAM_LBCn CLBCX = 2*"OPEN", CLBCY = 2*"OPEN" /

&NAM_CONF CCONF = "START", configuration START or RESTA
          CEQNSYS = "LHE", Equation system
          NMODEL = 1, NVERB = 5,  number of models / verbosity
          CEXP = "CTRL0", CSEG = "SEG01",

&NAM_DYN XSEGLEN = 10800., segment length (s)
         XASSELIN = 0.2, LCORIO = T,
         XALKTOP = 0.001, XALZBOT = 14000. /
          parameters for vertical relaxation
```

```
&NAM_BACKUP XBAK_TIME(1,1) = 3600,  backup files
            XBAK_TIME(1,2) = 7200, XBAK_TIME(1,3) = 10800/

&NAM_BUDGET CBUTYPE='CART', XBULEN=600., XBUWRI=1800.
 parameters      NBUIL=50, NBUIH=60, NBUJL=50,
 for             NBUJH=65, NBUKL=2, NBUKH=10,
 BUDGET         LBU_KCP=F, LBU_ICP=T, LBU_JCP=T /

&NAM_BU_RTH LBU_RTH=T, CBULIST_RTH(1)="ALL"  budget for θ

&NAM_LES /  LES diagnostics

&NAM_SERIES /  temporal series

&NAM_BLANKn /
```

Name of backup file

Backup files from the run are named :

CEXP.NMODEL.CSEG.00n

Example :

```
CTRL0.1.SEG01.001
CTRL0.1.SEG01.002
CTRL0.1.SEG01.003

CTRL0.1.SEG01.000
```

synchronous files

all the variables at a given time

time-series files

temporal series, budget....

Simulations with SURFEX

PRE_IDEA1.nam :

With an input PGD file

```
&NAM_GRn_PRE CSURF='EXTE' /

&NAM_REAL_PGD CPGD_FILE = 'REUNION_PGD_1km5',
              LREAD_ZS= T, LREAD_GROUND_PARAM= T /

&NAM_LUNITn CINIFILE = 'REUNION_IDEA',
            CINIFILEPGD = 'REUNION_PGD_1km5' /
```

Without an input PGD file : 2 files in output(initial + PGD)

```
&NAM_GRn_PRE CSURF='EXTE' /

&NAM_LUNITn CINIFILE = 'REUNION_IDEA',
            CINIFILEPGD = 'REUNION_PGD_1km5' /
```

EXSEG1.nam :

```
&NAM_LUNITn CINIFILE = "BILAN.1" /
```

initial file

```
            CINIFILEPGD = "REUNION_PGD_1km5" /
```

PGD file associated to the initial file